Name: SOLUTIONS Date: ______

Year 9 Mathematics

Test 10, 2015

Topics - Trigonometry

____ / 60
= ____ %

Total Time: *60 minutes*

Total Reading: *5 minutes*

Total Working: *55 minutes*

Weighting: ______*% of the year.*

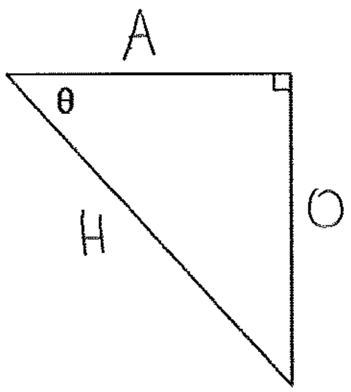
Equipment: *Curriculum Council Formula Sheet; ½ page notes (A4 one side), CAS calculator; Scientific Calculator*

CALCULATOR ALLOWED

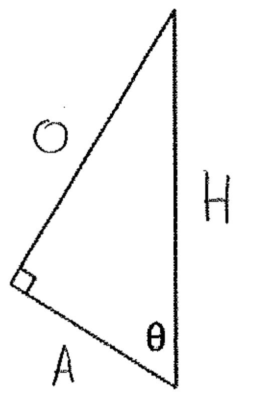
1. [2 marks: 1, 1]

Label the sides Hypotenuse, Opposite and Adjacent on the following right angled triangles:

a)



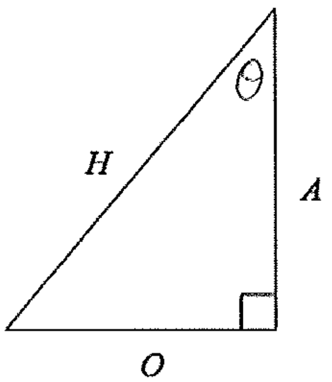
b)



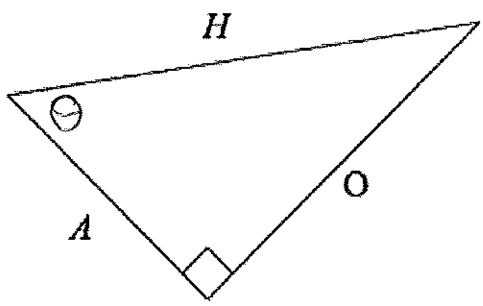
2. [2 marks: 1, 1]

Label the angle θ in the correct place on the following right angled triangles:

a)

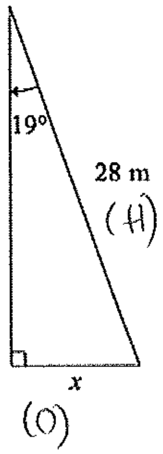


b)



3. **[8 marks: 2, 2, 2, 2]**
Calculate the length of the missing side for each right angled triangle:

a)

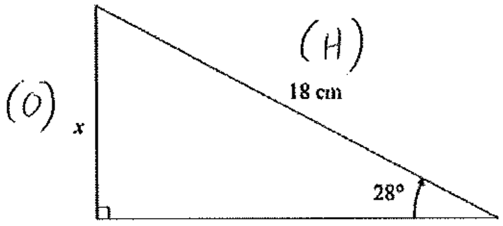


$\sin(19) = x/28$

$28 \times \sin(19) = x$

$9.12\text{ m} = x$

b)

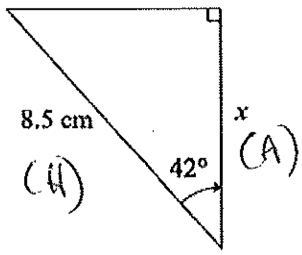


$\sin(28) = x/18$

$18 \times \sin(28) = x$

$8.45\text{ cm} = x$

c)

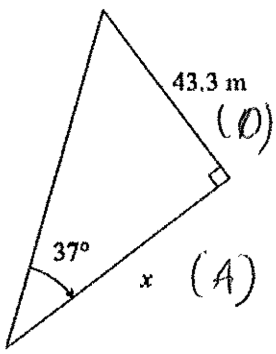


$\cos(42) = \frac{x}{8.5}$

$8.5 \times \cos(42) = x$

$6.32\text{ cm} = x$

d)



$\tan(37) = 43.3/x$

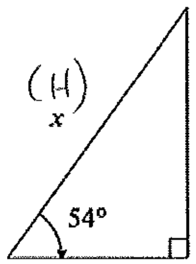
$x \times \tan(37) = 43.3$

$x = 43.3/\tan(37)$

$x = 57.46\text{ m}$

4. **[8 marks: 2, 2, 2, 2]**
Calculate the length of the missing side for each right angled triangle:

a)



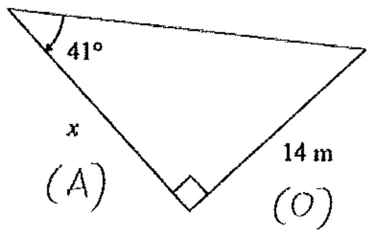
$\cos(54) = 4.2/x$

$x \times \cos(54) = 4.2$

$x = \frac{4.2}{\cos(54)}$

$= 7.15\text{m}$

b)



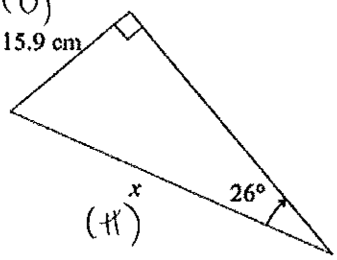
$\tan(41) = 14/x$

$x \times \tan(41) = 14$

$x = \frac{14}{\tan(41)}$

$= 16.11\text{ m}$

c)



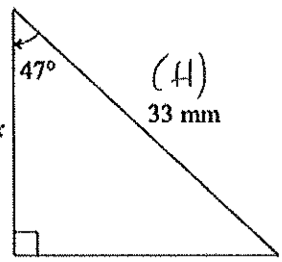
$\sin(26) = \frac{15.9}{x}$

$x \times \sin(26) = 15.9$

$x = \frac{15.9}{\sin(26)}$

$= 36.27\text{ cm}$

d)



$\cos(47) = x/33$

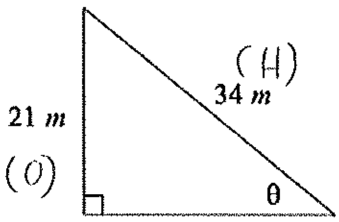
$33 \times \cos(47) = x$

$22.51\text{ mm} = x$

5. **[8 marks: 2, 2, 2, 2]**

Calculate the size of the missing angle for each right angled triangle:

a)

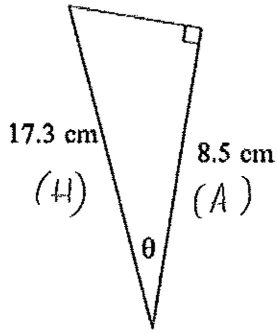


$\sin(\theta) = 21/34$

$\theta = \sin^{-1}(21/34)$

$= 38.1°$

b)

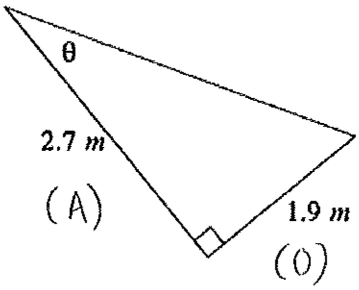


$\cos(\theta) = 8.5/17.3$

$\theta = \cos^{-1}(8.5/17.3)$

$= 60.6°$

c)

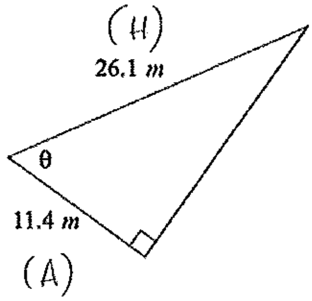


$\tan(\theta) = 1.9/2.7$

$\theta = \tan^{-1}(1.9/2.7)$

$= 35.1°$

d)



$\cos(\theta) = 11.4/26.1$

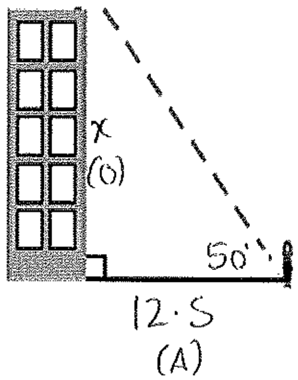
$\theta = \cos^{-1}(11.4/26.1)$

$= 64.1°$

For each worded question fill in the measurements on the given diagram.

6. **[9 marks: 3, 3, 3]**

a) A person stands 12.5 m from the base of a building, the angle of elevation to the top of the building is 50°. What is the height of the building?



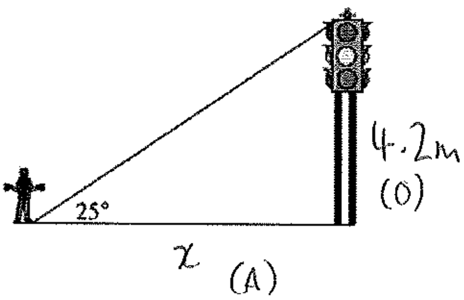
$$\tan(50) = \frac{x}{12.5}$$

$$12.5 \times \tan(50) = x$$

$$14.8969 = x$$

$\therefore$ Building is 14.9m tall

b) Find how far the person is standing from the 4.2m tall traffic lights.



$$\tan(25) = 4.2/x$$

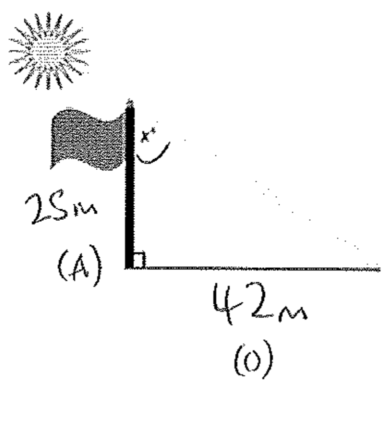
$$x \times \tan(25) = 4.2$$

$$x = 4.2/\tan(25)$$

$$= 9.01$$

$\therefore$ The person is 9m from the traffic light

b) A 25 m flagpole casts a 42 m shadow. What is the angle the sun makes with the flagpole



$$\tan(x) = 42/25$$

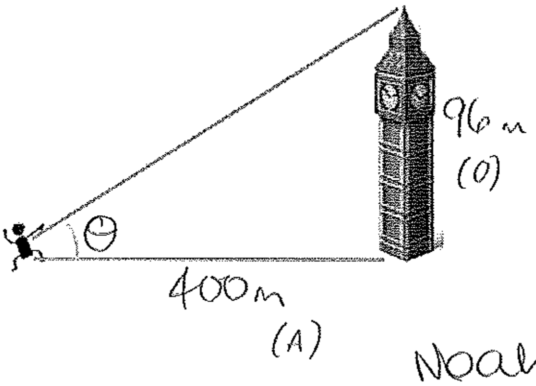
$$x = \tan^{-1}(42/25)$$

$$= 59.2^\circ$$

The sun makes a 59° angle with the pole.

7. **[6 marks: 3, 3]**

a) Noah sees the 96 m tall tower Big Ben from 400 m away. At what angle of elevation does he view it?



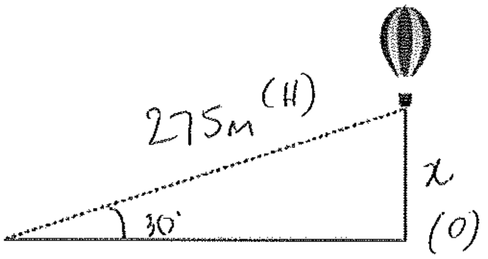
$\tan \theta = 96/400$

$\theta = \tan^{-1}(96/400)$

$= 13.5°$

Noah sees Big Ben @ 14° angle of elevation.

b) A hot air balloon is tethered to the ground by a a 275m rope which creates an angle to the ground of 30°. How high above the ground is the balloon?



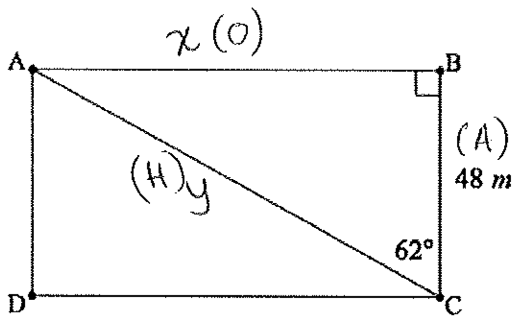
$\sin(30) = x/275$

$275 \times \sin(30) = x$

$137.5 = x$

The balloon is ≈ 138m above ground

8. **[5 marks]**
Jessie walked from A to B to C.
Linh walked from A to C.
How much further did Jessie walk?



Jessie $\tan(62) = x/48$

$48 \times \tan(62) = x$

$90.3m = x$

∴ Jessie walked $90.3 + 48 = 138.3m$

Linh $\cos(62) = 48/y$

$y \times \cos(62) = 48$

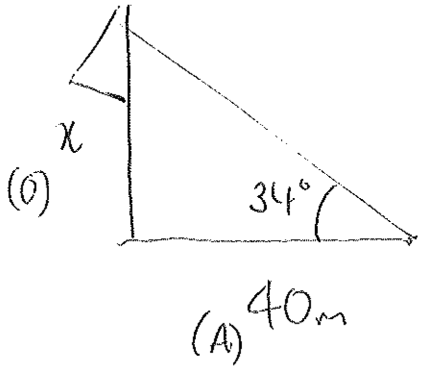
$y = 48/\cos(62)$

$= 102.2m$

∴ Jessie walked 36.1m further

9. **[6 marks: 3, 3]**

a) A flagpole stands on level ground. From a point 40 m from its base, the angle of elevation to the top of the pole is 34°. How tall is the flagpole?
(Sketch a diagram of the scenario)



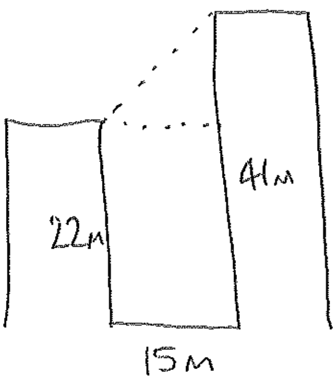
$\tan(34) = x/40$

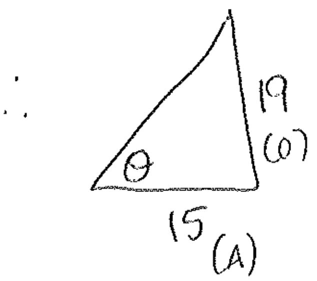
$40 \times \tan(34) = x$

$26.98 = x$

The flagpole is 27 m tall.

b) Two buildings are directly opposite one another either side of a 15 m wide street. One building is 22 m tall and the other is 41 m tall. Find the angle of elevation from the top of the shorter building to the top of the taller building.
(Sketch a diagram of the scenario).





$\tan(\theta) = 19/15$

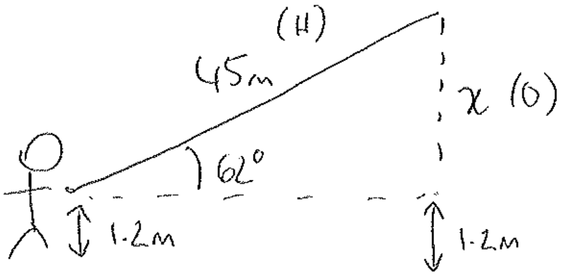
$\theta = \tan^{-1}(19/15)$

$= 51.71^\circ$

The angle of elevation to the top of the taller building is 52°

10. [6 marks: 3, 3]

a) A person flying a kite holds the string 1.2 m above level ground, at an angle of elevation of 62°. If the string is 45 m long, what is the height of the kite above ground level.
(Sketch a diagram of the scenario).



$\sin(62) = x/45$

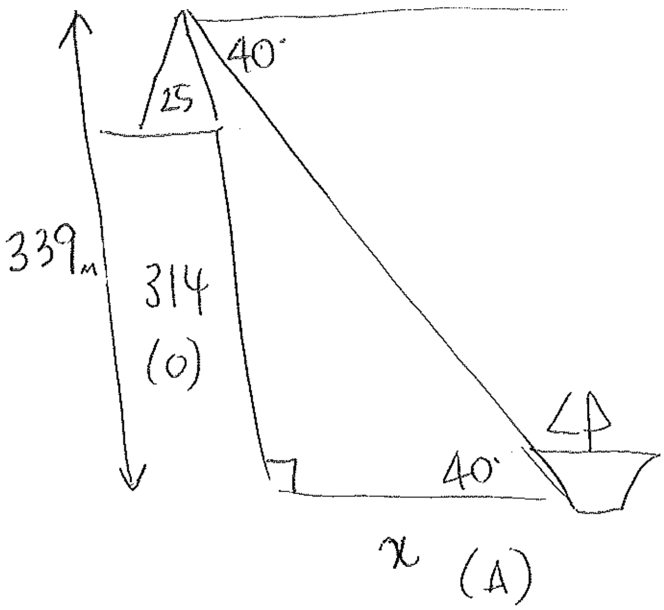
$45 \times \sin(62) = x$

$39.7\text{ m} = x$

∴ Height above ground

$39.7 + 1.2 = 40.9\text{m}$

b) From the top of a 25 m lighthouse, on a 314 m tall cliff, the angle of depression to a sailing boat out in the ocean is 40°. How far is the sailing boat from the base of the lighthouse?
(Sketch a diagram of the scenario).



$\tan(40) = 339/x$

$x \times \tan(40) = 339$

$x = 339/\tan(40)$

$= 404$

∴ The sail boat is 404 m from the cliff.

~ **END OF TEST** ~